Pranav Aggarwal

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.49
Modern Vidya Niketan, Sec-17, Faridabad	Class XII (CBSE)	2023	92.00%
Modern Delhi Public School, Sec-87, Faridabad	Class X (CBSE)	2021	96.80%

WORK EXPERIENCE

Agentic AI Intern -- DeepDrishti.ai (6 months)

Built production-grade RAG and LLM automation systems across patent intelligence and regulatory compliance, using GraphRAG, vector search, and multi-stage LLM validation. Cut manual review time by 40%+ and improved retrieval accuracy, delivering traceable, evidence-linked outputs with Docker-based deployment.

SOP Validator -- AI Compliance

- Built robust atomic-action extraction with staged LLM validation and GraphRAG evidence linking for traceable review
- Automated compliance validation across 250+ SOPs against internal guidelines and FDA warning-letter observations
- Cut validation time to ~10 min/SOP by staging applicability checks before deeper LLM-based compliance validation
- Reduced LLM inference calls by 40% while linking coverage decisions to SOP sections, gaps, and supporting evidence

PatentHub -- AI Patent Intelligence

- Engineered an Agentic Hybrid RAG + GraphRAG pipeline using Qdrant, PostgreSQL, and adaptive chunking
- Processed 12K+ patents (~50M tokens) with distributed, multi-stage technical evidence extraction
- Applied cross-encoder re-ranking and adaptive chunking, improving Top-K retrieval precision by 25%+
- Enabled multi-hop reasoning over patent corpora at 1.6s P95, replacing slow, manual patent analysis

PROJECTS

Limit Order Book & Matching Engine -- GitHub

- Designed a price-time priority matching engine in modern C++20, supporting Limit, Market, IOC & FOK orders daily.
- Built a hybrid map + list + hash-map design, cutting order cancellation time down to O(1) instead of O(n) scans.
- Added atomic Fill-or-Kill validation so orders execute fully or not at all, removing partial-fill risk entirely.
- Built a deterministic benchmark harness to accurately measure matching-engine throughput and order latency times.

Document-Based Question Answering System (RAG Chatbot) GitHub

- Built a RAG-based chatbot capable of answering user queries from uploaded PDF/TXT documents with >92% accuracy.
- Integrated LangChain and Sentence Transformers for efficient document embedding, indexing, semantic query matching.
- Processed and analyzed documents exceeding 200+ pages, enabling scalable and context-aware information retrieval.
- Enhanced document comprehension and accessibility by allowing users to extract insights without manual reading.

Loan Approval Prediction using XGBoost and Random Forest GitHub

- Built a classification model using Random Forest and XGBoost, with 92.6% and 91.5% accuracy with an ROC-AUC 0.97.
- Applied data preprocessing, SMOTE balancing, and GridSearchCV (162 fits) for optimized hyperparameter tuning.
- Conducted feature importance analysis and visualized ROC curves, performance comparisons for model interpretability.
- Utilized Python (Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib) for end-to-end model development and evaluation.

SKILLS

Languages & CS Core: C++ (STL, OOP), Python, C, SQL, Data Structures & Algorithms, OS, DBMS

Backend & Tools: PostgreSQL, Qdrant (vector DB), Hybrid Retrieval, Cross-Encoder Re-ranking, Docker, Git/GitHub

Gen AI & LLMs: RAG, GraphRAG, Sentence Transformers, LLM-based Validation, Agentic Pipelines

Machine Learning: scikit-learn, XGBoost, NumPy, Pandas, Matplotlib, Seaborn, SMOTE, GridSearchCV

Coding Profile: LeetCode: 250+ problems solved (26 Hard), 50-Day Badge 2025 & 2026 LeetCode

RELEVANT COURSES

Completed **Data Structures and Algorithms(DSA)** course on Udemy, focusing on problem solving using C++

Supervised Machine Learning: Regression and Classification || DeepLearning.AI and Stanford University || Coursera

CSE Electives: Introduction to Artificial Intelligence , Introduction to Programming , Machine Learning in Chemistry

Electives: Calculus , Probability, Elementary Linear Algebra, Introductory Economics, Macroeconomics

POSITIONS OF RESPONSIBILITY

Office of Career Services - Internship Coordinator

- Coordinated internship hiring drives for 300+ students, streamlining the entire application and selection process.
- Served as the Single Point of Contact (SPOC) for 50+ recruiting companies, enhancing institute-industry relations.

Alumni Student Relations Cell - Campus Tour Coordinator

- Organized 15+ guided campus tours for alumni and special guests, strengthening alumni-student engagement.
- Trained and led a team of 10+ volunteers, ensuring professional and well-coordinated campus experiences.

EXTRACURRICULAR

- National Service Scheme (NSS) Volunteer- Actively participated as NSS volunteer in the Swachhata campaign at IIT Hyderabad